

Stoquify Enterprise Agent and Skill System

Decision report | 15 July 2026

Scope: Repository-grounded assessment of the agents, reusable skills, architecture, controls, metrics, and delivery sequence required to make Stoquify an indispensable daily operating system for OHADA-oriented retail and distribution businesses.

Executive Summary

- **Build a controlled operating copilot, not a swarm of autonomous bots.** Stoquify already owns substantial operational truth across POS, inventory, purchasing, payments, reconciliation, accounting, close, compliance, HRIS, and payroll. Its fastest route to daily stickiness is a single role-aware Stoquify Command Agent that turns existing trusted snapshots and action queues into an explainable brief, then routes approved work through narrowly scoped skills.
- **The first production portfolio should contain three user-facing agents and one shared orchestration core.** Build the Stoquify Command Agent, Cash and Reconciliation Agent, and Inventory and Replenishment Agent. Build the Exception and Action Orchestrator as shared infrastructure rather than another chat persona. These use cases combine high frequency, strong existing data, measurable value, and defensible workflow depth.
- **Do not permit autonomous financial, payroll, compliance, entitlement, or destructive mutations in the first release.** Existing deterministic services must remain the source of truth. Agents may retrieve, explain, prioritize, draft, and propose. A permission check, evidence check, explicit approval, fresh authentication where required, and an idempotent service command must precede every high-impact write.
- **Close the control-plane and measurement gaps before broad rollout.** The repository has no production agent runtime or product-usage analytics layer. The module program also reports 58 active surface gaps and a no-go posture for broad enforcement. Agent execution, approval, cost, evaluation, and feedback records must be added before claiming enterprise readiness.

Direct recommendation: spend the first 90 days proving that a trusted command brief plus cash and inventory exception workflows increase action completion and reduce reconciliation or stock-resolution time for a small pilot cohort. Do not build a general-purpose autonomous agent platform, custom foundation model, autonomous accounting poster, automatic statutory filer, or employee-level payroll chatbot during this period.

1. Decision Frame and Definitions

This report supports a product and architecture decision by Stoquify leadership: which agent capabilities should be built, in what order, and under what controls so the platform becomes a daily operating habit without weakening financial or regulatory trust.

For this assessment:

- **Agent** means a bounded, stateful software actor that interprets a goal, selects approved skills, and may continue a workflow across steps. An agent is not the source of financial or operational truth.
- **Skill** means a narrow, versioned, testable capability with a defined input, output, permission boundary, evidence contract, and safe failure state.
- **Tool** means a deterministic application operation, normally an existing Stoquify service command or read model, that enforces the business rules.
- **Autonomy** means the ability to continue without a human at every step. It does not imply authority to bypass approval, RBAC, tenant scope, module entitlement, fresh authentication, or accounting controls.
- **Moat** means an advantage that compounds through proprietary workflow evidence, integrations, corrections, country expertise, and trusted operating history. The language model itself is not the moat.

2. Evidence Reviewed and Current-State Confidence

The assessment used current repository evidence, including:

- docs/architecture/system/ARCHITECTURE.md
- config/sidebar.ts and config/permissions.ts
- prisma/schema.prisma
- package.json
- graphify-out/GRAPH_REPORT.md
- services/daily-habit/daily-habit-digest.service.ts
- services/manager-action-center/manager-action-center.service.ts
- services/owner-war-room/owner-war-room.service.ts
- current readiness JSON reports for close truth, payment/cash truth, purchasing/AP, offline POS replay, statutory country packs, report trust, role cockpits, public-identity abuse, and CI
- the module control-plane status and its dated surface inventory
- the existing OHADA SMB skills audit and transaction-history agent-to-skill matrix
- official product documentation for Shopify Sidekick, Intuit Assist, Microsoft Dynamics 365 agents, Odoo AI agents/server actions, and Toast IQ.

The canonical architecture document says the platform has been live for roughly six months with about 50 paying organizations. That statement was last updated on 23 May 2026 and was not reconciled against a billing system in this assessment, so it is context rather than a current commercial metric.

What the repository proves

Stoquify is already a broad multi-tenant operating platform:

- 137 application page/API route files were counted.
- 43 service-domain directories and 37 action-domain directories are present.
- The Prisma schema contains operational and control models spanning tenant identity, inventory, POS, offline replay, purchasing/AP, payments, reconciliation, accounting, close assurance, compliance, HRIS, payroll, workflow assurance, and audit.
- The sidebar is intentionally grouped into Command, Operations, Finance and Trust, People, and Governance.
- Existing daily-digest, manager-action-center, and owner-war-room services already combine source-owned snapshots, freshness, evidence grades, permissions, redaction, signals, and action queues.
- Current dated readiness reports show all checks passing for nine narrow control areas, including payment/cash truth, close invalidation, purchasing/AP, offline POS replay, report trust, role cockpit, and CI.

What the repository does not yet prove

- A focused source scan found no production agent-run, tool-execution, prompt-version, model-provider, agent-memory, or agent-approval runtime in application code or the Prisma schema.
- A focused source scan found no product analytics system for activation, retention, feature usage, daily active use, weekly active use, or time to value.
- The module-control program remains no-go for broad enforcement. Its 14 July report records 323 surfaces, 58 active gaps, open P0 security prerequisites, mixed enforcement truth, and no rollout certification for any catalog module.
- Passing static readiness gates proves specific controls exist; it does not prove customer adoption, agent accuracy, workflow savings, or production economics.
- Statutory controls correctly block unsupported production claims. They do not establish that every country adapter is production-certified.

3. The Platform Is Agent-Ready in Data, but Not Yet in Runtime Governance

The strongest existing foundation is the flow from service-owned state to snapshots, evidence-aware signals, permission-filtered action queues, and role-specific workspaces. This is exactly the substrate a safe agent needs.

Capability	Current assessment	Evidence and implication
Service-owned domain truth	Strong	Services, source links, posting controls, and boundary gates provide safe tools.
Tenant/RBAC controls	Strong but incomplete	Central permissions and protected services exist; module-control P0 issues remain.
Evidence, freshness, and redaction	Strong	Existing BI contracts, snapshots, proof drawers, and export gates can ground answers.
Daily role workflows	Strong foundation	Daily Digest, Manager Action Center, and Owner War Room are natural agent entry points.
Cross-domain event and assurance model	Moderate to strong	Business events, outbox, assurance runs, incidents, and audit records exist.
Durable module entitlement truth	Incomplete	Current program explicitly blocks broad enforcement and privileged administration.
Product adoption telemetry	Missing	No reliable baseline exists for activation, repeat use, or value realization.
Agent runtime and approvals	Missing	No durable agent-run, step, tool-call, approval, feedback, or cost ledger was found.
Model governance and evaluations	Missing	No prompt/model registry, evaluation suite, or agent-specific incident controls were found.
Country-pack production coverage	Controlled but limited	The platform fails closed; production coverage must remain evidence-specific.

Implication: the lowest-effort path is to wrap existing read models and service commands with a thin agent control plane. Rebuilding domain logic in prompts, creating a parallel AI database, or allowing models to write directly to Prisma would destroy the platform's strongest advantage.

4. Daily and Hourly User Workflow Map

Person a	Daily/hourly job	Current Stoquify substrate	Agent opportunity	Human control required
Owner	Understand cash, stock exposure, supplier commitments, payroll exposure, and close risk	Owner War Room, tenant snapshots, proof subjects	Morning and exception brief with evidence-backed priorities	Approval for payments, sensitive exports, overrides, or policy changes
Manager	Clear overdue, critical, blocked, assigned, and routine actions	Manager Action Center and action queue	Continuous exception routing and next-best-action coaching	Approval for operational mutations outside delegated authority
Cashier/ POS operator	Sell quickly, handle exceptions, close sessions, and preserve receipt truth	POS, cash drawer, offline queue, receipt tokens	Contextual help, discrepancy explanation, and safe recovery guidance	Manager approval for voids, refunds, or overrides
Stockkeeper	Prevent stockouts, resolve negative stock, count, transfer, and investigate variance	Inventory levels/events/cou nts/transfers and stock signals	Replenishment proposals and variance investigation	Approval for adjustments, write-offs, and destructive corrections
Purchasing officer	Create POs, receive goods, resolve three-way match issues, and protect supplier payments	PO, goods receipt, supplier invoice, AP controls	Supplier commitment brief and variance-resolution workflow	Maker-checker approval for bank changes and payments
Finance/accountant	Reconcile payment rails, clear suspense, post safely, close periods, and export proof	Reconciliation , ledger, close assurance, report certification	Matching suggestions, exception explanation, close blocker resolution	Fresh auth and approval for posting, reversal, sign-off, certification, and export
HR/payroll operator	Prepare employee inputs, run payroll, handle payments and declarations	HRIS/payroll control services and proof chains	Readiness and variance agent with aggregate-safe explanations	Segregated approvals for compensation, payroll, payments, and declarations
Employee	Access payslips and resolve personal workflow issues	Payroll self-service	Narrow self-service guide grounded only in the employee's own records	No cross-employee access; sensitive changes require HR workflow
Platform administrator	Manage users, roles, modules, security, and integrations	Governance surfaces and module-control workbench	Configuration diagnostics and impact previews	No autonomous role, entitlement, secret, or tenant-policy changes

5. World-Class Pattern: Embedded, Contextual, Reviewable Action

Current official product documentation points toward a consistent pattern:

- Shopify Sidekick is available within the store context, can analyze and complete tasks, runs longer work in the background, remembers explicit context, supports saved skills, and presents changes for review before applying them.
- Microsoft Dynamics 365 connects finance agents to ERP context and treats agent execution as a governed, metered platform capability.
- Odoo separates AI decision-making from deterministic workers: the model selects a tool, while ordinary server logic enforces the business rules and performs the write.
- Toast IQ combines a daily briefing, proactive recommendations, plain-language analysis, and in-product action across connected operational data.
- Intuit Assist is positioned as a trusted financial assistant embedded in the small-business workflow rather than a detached chatbot.

Stoquify should adopt the pattern, not copy the products: **one embedded entry point, record/page context, proactive briefs, reusable skills, asynchronous work, review before change, and deterministic tools**. Its differentiator should be evidence-backed stock-to-cash-to-close workflows for OHADA businesses, including low-connectivity operations and controlled statutory honesty.

6. Recommended Agent Portfolio

Weighted scoring model

Each candidate was scored from 1 to 5. The weighted result is also on a 5-point scale.

Criterion	Weight	Interpretation
User value	18%	Magnitude of pain or outcome improved
Frequency	14%	Hourly/daily recurrence
Retention potential	12%	Ability to create a durable habit and switching cost
Strategic differentiation	11%	Fit with Stoquify's evidence and OHADA operating-system thesis
Revenue potential	8%	Upgrade, expansion, or pricing power
Data-moat potential	8%	Ability to compound proprietary workflow evidence
Time to measurable value	8%	Speed to a credible pilot result
Reusability	7%	Shared value across roles and modules
Implementation effort, inverse	5%	Higher score means lower relative effort
Dependency readiness	4%	Existing data, services, and controls
Operating cost, inverse	3%	Higher score means lower expected run cost
Security/compliance risk, inverse	2%	Higher score means lower residual risk

Rank	Agent	Score	Decision	Primary reason
1	Stoquify Command Agent	4.70	Build now	Creates one daily entry point from existing trusted briefs and actions.
2	Exception and Action Orchestrator	4.65	Build now - platform core	Reusable routing, approvals, asynchronous work, and escalation for every domain.
3	Cash and Reconciliation Agent	4.55	Build now	Daily pain, strong evidence base, direct financial value, and a defensible moat.
4	Inventory and Replenishment Agent	4.45	Build now	High-frequency operational value with strong existing stock/event models.
5	Close and Compliance Agent	4.20	Build next	High trust and revenue value, but higher regulatory and approval burden.
6	Purchasing and AP Agent	4.15	Build next	Strong three-way-match and supplier-control substrate; depends on clean supplier data.
7	Platform Assurance Agent	3.85	Build next - internal	Essential for safe scaling but not a direct customer-facing habit.
8	Customer Success and Adoption Agent	3.85	Build later	Valuable after product telemetry and playbooks exist.
9	Payroll and Workforce Agent	3.60	Build later, narrow pilot	Valuable but sensitive, country-dependent, and expensive to validate safely.

6.1 Stoquify Command Agent

Class: Advisory and workflow agent. **Users:** owners, managers, accountants, purchasing teams, and stockkeepers.

Job: answer "What needs my attention now, why, and what is the safest next action?" from any authenticated page. It should generate role-specific morning, shift, and end-of-day briefs and continue approved workflows in the background.

Triggers: user request; login/daily digest; critical or high business signal; scheduled morning/end-of-day brief; completion or failure of a delegated workflow.

Inputs and tools: tenant operating snapshot, payment-truth snapshot, inventory-cash snapshot, close-readiness snapshot, action queue, evidence/proof service, current route/record context, role and permission context, module entitlements, organization locale/currency, and approved domain skills.

Outputs: concise brief; ranked actions; evidence links; explanation of stale, unavailable, redacted, or permission-denied state; proposed next step; notification when a background task requires review.

Autonomy: read and analyze automatically. Draft low-risk changes. Never execute a high-impact write without policy evaluation and approval.

MVP: read-only brief, evidence-grounded Q&A, action routing, and deep links. **Mature scope:** approved low-risk actions, saved user routines, voice, multilingual assistance, and asynchronous task completion. **Exclude:** arbitrary database access, generic web answers presented as tenant truth, and autonomous financial decisions.

Success metrics: weekly active eligible users, brief-to-action conversion, action completion within SLA, evidence-link use, material correction rate, and cost per completed action.

6.2 Exception and Action Orchestrator

Class: Orchestration and monitoring agent. **Users:** all domain agents and internal operations.

Job: convert signals into deduplicated cases, assign them to a role, invoke the right skill, manage deadlines and approvals, and close the loop when evidence changes.

Triggers: business signal, assurance incident, scheduled rule evaluation, user delegation, tool completion, timeout, rejected approval, or changed source evidence.

Inputs and tools: signal rules, action queue, workflow assurance, notification service, approval service, agent-run ledger, idempotency keys, and escalation policy.

Outputs: case/run record, owner, due time, state, evidence fingerprint, approval request, retry/escalation event, and terminal outcome.

Autonomy: may prioritize, assign within policy, notify, pause, retry safe reads, and request approval. It may not override a denial, change permissions, suppress evidence, or retry non-idempotent writes automatically.

MVP: shared run state and routing for the first three agents. **Mature scope:** cross-domain workflow plans and service-level objectives. **Exclude:** user-facing personality and business-rule ownership.

Success metrics: duplicate-case rate, mean time to assignment, completion within SLA, stuck-run rate, retry recovery rate, and unauthorized-tool-call rate.

6.3 Cash and Reconciliation Agent

Class: Monitoring, advisory, assurance, and approval-gated workflow agent. **Users:** finance officers, accountants, owners, and controllers.

Job: explain cash position and payment-truth exceptions, propose statement matches, prioritize suspense, identify missing evidence, prepare a reconciliation pack, and guide close-safe resolution.

Triggers: statement import, provider-event drift, reconciliation run completion, new suspense/exception, approaching close, or user request.

Inputs and tools: provider accounts, statement lines, payment transactions, match records, suspense items, exceptions, source manifests, ledger source links, period state, close blockers, and reconciliation services.

Outputs: ranked exceptions, matching suggestions with confidence and rationale, evidence gaps, proposed resolution path, and certificate/readiness preview.

Autonomy: may auto-classify and suggest. Exact deterministic rules may auto-match within existing policy. Model-generated matches require review until precision is proven. Posting, suspense resolution, sign-off, and certification require the existing permission, maker-checker, and fresh-auth controls.

MVP: explain and prioritize unmatched/suspense items; suggest matches; prepare approval packets. **Mature scope:** configurable confidence bands and proactive cash-risk forecasting. **Exclude:** moving funds, initiating credit, creating providers, or overriding period locks.

Success metrics: median exception-resolution time, suggestion acceptance rate, precision of accepted suggestions, reduction in aged suspense, close-blocker reduction, and zero unauthorized postings.

6.4 Inventory and Replenishment Agent

Class: Monitoring, advisory, and workflow agent. **Users:** owners, managers, stockkeepers, and purchasing teams.

Job: detect stockout and overstock risk, explain negative or unusual inventory, recommend replenishment or transfer, and guide variance investigations using event history.

Triggers: stock threshold, sales velocity change, count variance, negative stock, delayed PO, transfer conflict, stale projection, or user request.

Inputs and tools: inventory levels/events, stock counts, transfers, item/supplier records, sales velocity, PO commitments, location scope, inventory valuation, and event/reconciliation services.

Outputs: reorder/transfer proposal, root-cause trace, affected locations/items, cash exposure, evidence links, and proposed investigation checklist.

Autonomy: may monitor, explain, and draft POs/transfers. Creating or approving a PO, posting an adjustment, write-off, or valuation correction remains approval-gated.

MVP: top stock risks, reorder suggestions, and negative-stock root-cause trace. **Mature scope:** service-level-aware demand planning and multi-location balancing. **Exclude:** black-box demand forecasts without confidence intervals and autonomous write-offs.

Success metrics: stockout days, emergency purchase rate, accepted recommendation rate, count-variance resolution time, inventory cash tied up, and forecast error by item class.

6.5 Purchasing and AP Agent

Class: Workflow and assurance agent. **Users:** purchasing officers, managers, AP staff, and controllers.

Job: summarize supplier commitments, identify PO/receipt/invoice variance, prepare three-way-match resolutions, and surface payment or bank-change risk.

Guardrails: never approve its own proposal; never change supplier payment destinations; never release payment. Maker-checker, evidence, and fresh-auth controls remain deterministic.

MVP: variance explanation and action packet. **Decision:** build after the shared orchestrator and clean supplier/invoice data are proven in the pilot.

6.6 Close and Compliance Agent

Class: Assurance and workflow agent. **Users:** accountants, controllers, compliance officers, and external reviewers with scoped access.

Job: explain close blockers, assemble evidence, propose remediation order, track waivers/reviews, and state the exact production support level of the relevant country pack.

Guardrails: no autonomous posting, reversal, period close/reopen, certification, waiver approval, or statutory submission. The agent must quote the country-pack provenance and support state in every statutory recommendation.

MVP: close-blocker navigator and evidence checklist. **Decision:** build next, after agent evidence and approval primitives are stable.

6.7 Payroll and Workforce Agent

Class: Advisory and approval-gated workflow agent. **Users:** payroll operators, HR managers, finance controllers, and employees through a separate self-service boundary.

Job: identify input-readiness gaps, explain aggregate payroll variance, prepare review steps, and answer employee-specific self-service questions without exposing another employee's data.

Guardrails: no person-level payroll data in owner briefs or general memory; no autonomous compensation, bank-destination, run approval, posting, payment, or declaration action.

MVP: read-only readiness and aggregate variance for one reviewed country pack. **Decision:** build later because privacy, country rules, and approval requirements make validation costly.

6.8 Customer Success and Adoption Agent

Class: Advisory and workflow agent. **Users:** customer administrators, champions, and Stoquify customer-success staff.

Job: guide onboarding, detect unused critical workflows, recommend training, and create role-specific adoption plans.

Dependency: product telemetry, consented usage data, playbook versioning, and customer-health definitions. **Decision:** build later; without telemetry it would guess.

6.9 Platform Assurance Agent

Class: Internal assurance agent. **Users:** engineering, security, support, and release managers.

Job: monitor agent evaluations, policy denials, prompt/model changes, cost, latency, incidents, and stale evidence; suspend unsafe agent versions and assemble release evidence.

Guardrails: may suspend or fail closed, but may not silently relax policy or self-certify. **Decision:** build with the second production wave, before broad tenant rollout.

7. Reusable Skill Catalogue

Foundation and control skills

ID	Skill	Responsibility and safe failure	Build
S01	Trusted Context Resolver	Resolve tenant, user, role, location, route/record, locale, currency, period, and correlation context. Fail closed on ambiguity.	Now
S02	Permission and Entitlement Guard	Evaluate RBAC, tenant scope, module entitlement, action risk, and fresh-auth requirement. Return denial, never a workaround.	Now
S03	Evidence-Grounded Retrieval	Retrieve only approved service/read-model evidence with source identity, freshness, grade, and redaction. Return unavailable when unsupported.	Now
S04	Safe Action Planner	Convert a request into bounded steps, required tools, approvals, evidence, and rollback/compensation notes. Produce a proposal only.	Now
S05	Approval and Step-Up Coordinator	Create an approval request, enforce maker-checker separation, invoke verified step-up authentication, and record outcome. Pause safely.	Now
S06	Idempotent Tool Executor	Invoke allowlisted service commands with schema validation, idempotency key, timeout, and deterministic error mapping. Never call Prisma directly.	Now
S07	Agent Evidence Recorder	Persist prompt/model/skill/tool versions, inputs by reference, evidence fingerprints, decisions, approvals, outputs, and terminal status.	Now
S08	Redaction and Disclosure Policy	Apply field- and role-level disclosure rules before model input and user output. Fail closed on unknown sensitive fields.	Now
S09	Freshness and Trust Evaluator	Interpret snapshot age, evidence grade, blockers, redactions, certification state, and unsupported claims.	Now
S10	Exception Prioritizer	Deduplicate and rank cases by severity, value, deadline, evidence quality, and role. Avoid inventing urgency.	Now
S11	Notification and Escalation Router	Route approval, deadline, failure, and completion notices through user preferences and role policy.	Now
S12	Agent Run State Machine	Manage queued, running, waiting-for-approval, blocked, retryable, failed, cancelled, and completed states with resumability.	Now

ID	Skill	Responsibility and safe failure	Build
S13	Model and Cost Router	Select an approved model by task sensitivity and complexity, enforce token/cost budgets, and fall back safely.	Next
S14	Explicit Preference Memory	Store user-approved preferences and saved routines with expiry and deletion. Never store raw payroll or secret material.	Next
S15	Offline and Replay Awareness	Detect provisional/offline state, avoid finality claims, and wait for server replay/certification before consequential action.	Next
S16	Country-Pack Provenance Resolver	Resolve country, effective date, version, review evidence, environment, and production-support state. Block unsupported automation.	Next

Domain skills

ID	Skill	Responsibility and safe failure	Build
S17	Daily Operating Brief	Compose role-specific, evidence-backed morning/shift/end-of-day briefs from existing snapshots and actions.	Now
S18	Cross-Domain Root-Cause Trace	Trace a signal through event, source record, ledger link, evidence, and related workflow without inferring missing links.	Now
S19	Cash Exception Triage	Classify and rank payment exceptions and suspense, showing evidence gaps and close impact.	Now
S20	Reconciliation Match Suggestion	Suggest statement/payment matches with confidence, features, rationale, and review requirement.	Now
S21	Inventory Risk and Replenishment	Estimate stockout/overstock risk and propose reorder or transfer using location and supplier evidence.	Now
S22	Inventory Variance Investigation	Explain negative stock, count variance, replay conflicts, and projection drift from the event trail.	Now
S23	PO/Receipt/Invoice Variance	Explain three-way-match differences and prepare a controlled resolution packet.	Next
S24	Supplier Commitment and Payment Risk	Summarize expected cash commitments and supplier-control exceptions without releasing funds.	Next
S25	Close Blocker Navigator	Rank close findings and missing evidence; propose remediation sequence and show certification consequences.	Next
S26	Compliance Readiness Explanation	Explain document/submission state and country-pack support without making legal or production claims beyond evidence.	Next
S27	Payroll Readiness and Variance	Check input readiness and explain aggregate variance within a reviewed country pack and privacy boundary.	Later
S28	Adoption and Onboarding Coach	Convert product telemetry into role-specific setup, training, and adoption actions. Return unknown until telemetry is reliable.	Later

Every skill must have a versioned JSON input/output schema, permission and data classification, deterministic preconditions, maximum runtime/cost, retry class, evidence contract, audit record, evaluation cases, and explicit list of prohibited actions.

8. Agent-to-Skill Responsibility Matrix

Agent	Primary skills	Shared/secondary skills
Stoquify Command	S17 Daily Brief, S03 Retrieval, S04 Planner, S18 Root-Cause	S01-S02, S07-S11, S13-S16
Exception and Action Orchestrator	S10 Prioritizer, S11 Notifications, S12 Run State	S01-S09, S13, S15-S16
Cash and Reconciliation	S19 Cash Triage, S20 Match Suggestion, S18 Root-Cause	S01-S12, S13, S16, S25
Inventory and Replenishment	S21 Replenishment, S22 Variance, S18 Root-Cause	S01-S12, S13, S15, S23-S24
Purchasing and AP	S23 Three-Way Variance, S24 Commitment Risk	S01-S12, S13, S16, S18-S20
Close and Compliance	S25 Close Navigator, S26 Compliance Readiness, S16 Country Pack	S01-S13, S18-S20, S23-S24
Payroll and Workforce	S27 Payroll Readiness, S16 Country Pack	S01-S14, S18, S25-S26
Customer Success and Adoption	S28 Adoption Coach, S17 Daily Brief	S01-S04, S07-S14
Platform Assurance	S07 Evidence, S09 Trust, S10 Prioritizer, S12 Run State, S13 Model Router	All skill evaluation and policy metadata; no domain writes

9. Target Architecture and Dependency Map

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Authenticated user / scheduled trigger / business event
|
Agent experience
(Command, Cash, Inventory, later domains)
|
Exception & Action Orchestrator
run state | budget | routing | approvals | retry
|
Versioned skill registry
context | evidence | planning | domain reasoning | output
|
Policy and deterministic tool gateway
tenant + RBAC + entitlement + fresh auth + schema + idempotency
|
Existing Stoquify services and read models - source of truth
snapshots | signals | POS | inventory | purchasing | payments
accounting | close | compliance | HRIS/payroll | notifications
|
Postgres + business events/outbox + audit/evidence + observability

```

New control-plane records

Add a small, service-owned agent schema rather than a separate application:

- **AgentDefinition**: stable agent ID, purpose, owner, risk class, allowed skills, status, and default budget.
- **AgentVersion**: prompt, model policy, release state, evaluation result, effective period, and rollback target.
- **SkillDefinition** and **SkillVersion**: schemas, allowed tools, data classification, timeout, retry policy, and tests.
- **AgentRun**: tenant, actor, trigger, context references, status, budget, correlation ID, and terminal outcome.
- **AgentStep**: selected skill, evidence references, model/tool decision, structured output, and timing.

- **AgentToolCall**: allowlisted tool, validated arguments, idempotency key, result reference, and error class.
- **AgentApproval**: proposed action, approver policy, maker-checker evidence, fresh-auth result, decision, and expiry.
- **AgentEvidenceLink**: source type/ID, snapshot fingerprint, freshness, grade, redaction, and claim association.
- **AgentFeedback**: accepted, corrected, rejected, reason, and resulting workflow outcome.
- **AgentCostRecord**: model, tokens, latency, provider cost, retries, and cost allocation.
- **AgentIncident**: policy breach, unsafe output, evidence failure, prompt injection, excessive cost, or tool failure.

Raw sensitive records should not be copied into agent tables. Store references, hashes, classifications, and the minimum reviewable output.

Execution contract

1. Resolve tenant, actor, role, location, module, period, and route context on the server.
2. Retrieve only permission-filtered, redacted, evidence-graded service outputs.
3. Ask the model for a strict structured decision or plan, never a direct database mutation.
4. Validate the output against a versioned schema and policy.
5. For a read, return the answer with evidence and uncertainty.
6. For a write, create a proposed action and approval request.
7. After authorization and fresh auth, invoke an allowlisted service command with an idempotency key.
8. Persist the evidence, policy decision, approval, tool result, cost, and outcome.
9. Re-evaluate if evidence changed before execution; cancel stale proposals.

Model strategy

- Use a provider-neutral gateway and start with one primary and one fallback model.
- Use smaller models or deterministic logic for classification, summarization, and routing; use a stronger model only for complex cross-domain explanation.
- Keep prompts, model choice, temperature, output schema, and tool allowlist versioned.
- Do not fine-tune initially. First collect consented corrections and verified outcomes.
- Never send secrets, raw tokens, unrestricted payroll records, or other tenants' information to a model.

10. Autonomy and Human-Approval Policy

Risk tier	Examples	Agent authority
Tier 0 - read only	Explain a metric, summarize a case, show evidence	Automatic after permission and disclosure checks
Tier 1 - reversible draft	Draft PO, prepare reconciliation proposal, compose reminder	Agent may draft; user reviews before persistence or sending
Tier 2 - controlled operational write	Assign case, update non-financial status, create approved task	Explicit confirmation; deterministic service validation; full audit
Tier 3 - financial/sensitive write	Post/reverse journal, approve PO, stock adjustment, payroll run, payment action	Permission + maker-checker where applicable + fresh auth + explicit approval
Tier 4 - regulated/irreversible/admin	Period close/reopen, statutory submission, entitlement/RBAC change, secrets, destructive action	Agent may only explain and prepare evidence; authorized human executes controlled workflow

No agent may grant itself access, choose a less restrictive tool after denial, hide a blocker, mark unsupported country logic as production-ready, or treat conversation memory as authorization.

11. First Three Production Specifications

Specification A: Stoquify Command Agent MVP

Entry point: persistent copilot panel available from Daily Digest, Manager Action Center, Owner War Room, and evidence-aware domain pages.

Initial intents: `brief_me`, `explain_metric`, `why_blocked`, `show_evidence`, `prioritize_actions`, `route_to_workflow`, and `save_read_only_routine`.

Required skills: S01-S04, S07-S12, S17-S18. **Allowed tools:** read-only snapshot, action-queue, evidence, route-link, and notification-preference tools.

Acceptance criteria:

1. Every numeric or status claim links to an authorized evidence source and freshness state.
2. Permission-denied, redacted, stale, partial, empty, and unavailable states remain visible.
3. A user cannot retrieve another tenant, role, location, or employee's restricted information through conversation.
4. Unsupported requests produce a bounded refusal and route to the appropriate existing workflow.
5. Evaluation passes include English and French prompts, ambiguous requests, prompt injection, stale evidence, and denied tools.

Specification B: Cash and Reconciliation Agent MVP

Entry point: reconciliation workbench and Cash Command.

Initial intents: `summarize_cash_risk`, `explain_exception`, `suggest_match`, `show_missing_evidence`, `prepare_resolution`, and `prepare_close_review`.

Required skills: S01-S12, S18-S20, and S25. **Allowed tools:** reconciliation read models, statement/match evidence, suspense workflow proposal, period-state read, and approval creation. Posting tools are disabled in the MVP.

Acceptance criteria:

1. Match suggestions expose confidence, rationale, compared fields, and contradictory evidence.
2. Low-confidence or conflicting candidates are never presented as a resolved match.
3. Suggestions are tested against labeled historical cases and an untouched evaluation set.
4. The agent cannot sign, certify, post suspense, reverse, or bypass a closed period.
5. Accepted and rejected suggestions create feedback linked to the eventual outcome.

Specification C: Inventory and Replenishment Agent MVP

Entry point: inventory overview, movements, item detail, and manager action center.

Initial intents: `show_stock_risk`, `explain_negative_stock`, `recommend_reorder`, `recommend_transfer`, `investigate_variance`, and `draft_purchase_order`.

Required skills: S01-S12, S15, S18, S21-S24. **Allowed tools:** location-scoped inventory reads, sales velocity, supplier/PO commitments, event history, count/transfer state, and draft proposal creation.

Acceptance criteria:

1. Recommendations are location-specific and distinguish on-hand, available, committed, in-transit, and provisional/offline state.
2. Every recommendation includes demand window, confidence, supplier/lead-time assumption, and cash impact.
3. The agent never posts an adjustment, accepts a count, approves a PO, or writes off stock autonomously.

4. Forecast error and recommendation outcome are measured by item class and location.
5. Offline events remain provisional until replay/certification is complete.

12. Build Versus Buy

Capability	Recommendation	Reason
Foundation models	Buy through a provider-neutral gateway	Base-model training is not a Stoquify moat and would consume disproportionate capital.
Agent control plane	Build thinly inside Stoquify	Tenant, RBAC, evidence, approval, and domain integration are product-specific.
Domain skills and tool wrappers	Build	They encode Stoquify workflows, controls, OHADA context, and differentiated evidence.
Vector retrieval	Start with Postgres/approved managed retrieval	Avoid a separate data platform until document volume or latency proves the need.
Tracing and model observability	Buy or extend OpenTelemetry/Sentry-compatible tooling	Faster time to production; preserve canonical run/evidence records in Stoquify.
Product analytics	Buy a mature privacy-conscious event platform or build a minimal event layer first	The immediate need is reliable adoption and workflow telemetry, not custom analytics infrastructure.
Evaluation harness	Build domain datasets and assertions; use commodity runners	Proprietary cases and outcome labels are the moat; runner infrastructure is not.
Country-pack legal/accounting review	Partner with qualified experts	The platform can encode provenance and controls but must not self-certify law.
Voice and translation	Buy initially	Useful experience layer, but not a core differentiator.

13. Moat Strategy

13.1 Proof graph

Every recommendation should connect business signal -> source record -> workflow event -> ledger/evidence link -> approval -> final outcome. Competitors can imitate a chat UI; reproducing years of trustworthy, queryable operating proof is harder.

13.2 Outcome-labelled workflow graph

Capture which exception occurred, which recommendation was shown, which evidence changed the decision, who approved it, what tool executed, and whether the business outcome improved. This creates proprietary training and evaluation material without making the model the source of truth.

13.3 OHADA and country-pack provenance

Build versioned, effective-dated, expert-reviewed country packs linked to actual workflows. The moat is not a generic "Africa compliance" claim; it is controlled coverage with explicit supported, sandbox, blocked, and superseded states.

13.4 Stock-to-cash-to-close continuity

Stoquify can connect a sale, payment, inventory movement, supplier commitment, payroll exposure, reconciliation exception, ledger posting, and close blocker in one tenant-safe platform. Cross-domain resolution is more defensible than isolated AI features.

13.5 Low-connectivity operating knowledge

Offline devices, provisional receipts, replay evidence, conflict resolution, and server certification can become a differentiated operating dataset. Agents must understand provisional state rather than pretending offline data is final.

13.6 Embedded habits and switching cost

Daily briefs, saved routines, approval history, exception ownership, evidence packs, and accountant/customer collaboration create legitimate switching cost because the system becomes the business's operating memory.

What is not a moat

- A generic chatbot.
- A prompt library without workflow integration.
- A branded wrapper around a public model.
- Autonomous actions without evidence or governance.
- Static dashboards competitors can reproduce.
- Unsupported claims of legal, tax, or payroll automation.

14. Measurement Framework

No current product-usage baseline was found, so the following are **pilot exit thresholds**, not forecasts.

Primary outcome KPIs

1. **Agent-assisted action completion:** eligible high-priority actions completed within their SLA after an agent brief, divided by eligible surfaced actions. Pilot exit threshold: at least 40% with no degradation in control quality.
2. **Weekly value-active users:** eligible users who complete at least one evidence view, accepted recommendation, or approved agent-assisted workflow in a week, divided by eligible pilot users. Pilot exit threshold: at least 50% by week eight.
3. **Verified time saved:** median measured or sampled time saved per completed target workflow relative to the pre-pilot baseline. Pilot exit threshold: at least 20% for the selected cash or inventory workflow.

Drivers

- Seven-day agent activation.
- Brief opened -> evidence viewed -> action started -> action completed funnel.
- Recommendation acceptance and rejection reason.
- Mean time to assignment and resolution.
- Saved-routine repeat use.
- Evidence freshness and unavailable-answer rate.

Quality and trust guardrails

- Material correction rate below 10% for advisory outputs during pilot.
- Accepted match-suggestion precision at or above 95% before any confidence-based automation is considered.
- Zero cross-tenant, permission, entitlement, payroll-disclosure, or unauthorized-write incidents.
- 100% of consequential claims linked to authorized evidence with freshness and version metadata.
- 100% of Tier 3/4 attempts denied or routed to the required human-controlled workflow.
- Tool execution success at or above 99% for enabled idempotent actions, excluding valid business denials.
- Median interactive response latency below five seconds for briefs already backed by existing snapshots; asynchronous handling for longer work.
- Cost per successful outcome measured and capped by tenant/package. Do not optimize token cost at the expense of control quality.

15. Delivery Roadmap

Days 0-30: Foundation and measurement

- Freeze the agent/skill vocabulary, risk tiers, tool contract, and prohibited-action list.
- Add product telemetry for the current Daily Digest, action queue, reconciliation, and inventory workflows to establish baselines.
- Implement `AgentDefinition`, version, run, step, evidence, approval, feedback, cost, and incident records.
- Build S01-S12 and read-only tool wrappers around existing services.
- Create an evaluation corpus from synthetic cases plus reviewed, de-identified historical patterns.
- Resolve or explicitly isolate module-control PO gaps for the pilot routes; do not enable broad enforcement.
- Deliver the read-only Stoquify Command Agent behind tenant and user feature flags.

30-day gate: no write tools; all answers evidence-linked; denial and redaction tests pass; pilot telemetry is collecting; agent version can be suspended instantly.

Days 31-60: First controlled pilot

- Pilot the Command Agent and Exception/Action Orchestrator with 5-10 consenting organizations or a similarly bounded cohort.
- Add S17-S22 for daily briefs, cash triage/match suggestion, and inventory risk/variance.
- Enable background read tasks and review notifications.
- Establish weekly accuracy, adoption, latency, cost, and incident review.
- Label historical reconciliation outcomes and inventory exception outcomes for evaluation.

60-day gate: weekly value-active use and time-to-resolution show directional improvement; no material policy breach; correction and evidence-unavailable rates are understood.

Days 61-90: Approval-enabled low-risk workflows

- Add Tier 1 drafting: reconciliation resolution packet, draft reorder/transfer, and draft PO.
- Introduce explicit approval and stale-evidence revalidation.
- Add outcome feedback and saved read-only routines.
- Decide whether cash or inventory merits wider rollout based on measured value, not enthusiasm.
- Produce a production-readiness report with cost, security, accuracy, and rollback evidence.

90-day gate: expand only an agent whose primary KPI, trust guardrails, cost envelope, and support runbook pass. Keep other agents in pilot or stop them.

Months 4-6: Workflow expansion

- Add Purchasing/AP and Close Blocker skills.
- Introduce low-risk Tier 2 operational writes only after independent security and control review.
- Add English/French evaluation parity and country-pack provenance in responses.
- Build the Platform Assurance Agent and formal agent incident drills.

Months 7-9: Finance and compliance depth

- Expand cash/reconciliation to more providers and statement formats.
- Pilot Close and Compliance Agent in read-only/evidence-preparation mode.
- Introduce accountant collaboration and certified agent evidence packs.
- Add package-based usage limits and cost controls.

Months 10-12: Carefully earned autonomy and ecosystem

- Consider confidence-banded automation only for deterministic, reversible, low-risk cases with measured precision.
- Add Customer Success/Adoption Agent after telemetry definitions are stable.
- Pilot Payroll Readiness only in a reviewed country scope and aggregate-safe mode.
- Open a versioned integration/tool SDK only after tenant, approval, and audit boundaries are independently validated.

16. 30/60/90 Ownership and Acceptance Plan

Period	Accountable owner	Main deliverables	Acceptance evidence
0-30	Platform architect + security lead + product analytics owner	Control-plane schema, policy gateway, telemetry, S01-S12, read-only Command MVP	Threat model, schema/contract tests, cross-tenant negative tests, evaluation baseline, feature-flag rollback
31-60	Product lead + finance/inventory domain owners	Bounded pilot, S17-S22, background reads, reviewed datasets	Adoption funnel, accuracy report, latency/cost report, incident review, user interviews
61-90	Product lead + controller + operations lead	Tier 1 drafts, approvals, feedback loop, rollout decision	Primary KPI movement, precision/correction results, approval audit, support runbook, go/no-go decision

17. Risk Register and Release Gates

Risk	Impact	Mitigation	Release gate
Cross-tenant or over-privileged retrieval	Critical	Server-derived context, S02 guard, redaction before model input, adversarial tests	Zero leakage in negative and prompt-injection tests
Hallucinated financial or statutory claim	Critical	Evidence-only claims, source/freshness metadata, unsupported-state refusal	100% consequential claims traceable to evidence
Unauthorized or duplicate write	Critical	Approval policy, fresh auth, idempotency, deterministic service tool, stale-plan cancellation	Zero bypass; duplicate/replay tests pass
Prompt injection through records/documents	High	Treat retrieved content as data, tool allowlist, instruction isolation, output schema	Injection corpus produces no policy/tool escape
Sensitive payroll/privacy leakage	Critical	Data classification, aggregate-safe views, role-specific agents, memory exclusion	Disclosure tests and redaction audit pass
Country-rule drift	High	Effective-dated country packs, expert review, provenance resolver	Unsupported or stale pack blocks automation
Weak suggestions create operational noise	High	Confidence, outcome labels, quiet thresholds, user feedback	False critical alert rate and rejection reasons within limits
Model/provider outage	Medium	Fallback model, cached deterministic briefs, asynchronous retry, graceful unavailable state	Failure drill and no corrupted workflow state
Excessive cost/latency	Medium	Model routing, budgets, snapshot-first reads, asynchronous long tasks	Package cost cap and latency SLO met
Automation bias	High	Rationale, contradictory evidence, review, no false certainty	User testing shows comprehension; material correction tracked
Dirty or stale source data	High	Freshness evaluator, blocker visibility, no action on stale evidence	Stale-source simulations block or downgrade actions
Uncontrolled prompt/model change	High	Version registry, evaluations, canary, rollback, assurance agent	Candidate version passes evaluation and rollback drill
Module-control inconsistency	High	Pilot allowlist, resolve P0 prerequisites, no broad enforcement claim	Pilot surfaces have reconciled entitlement and permission truth

18. Recommendations by Decision Class

Build now

- Thin agent control plane and product telemetry.
- Foundation skills S01-S12.
- Stoquify Command Agent in read-only mode.
- Exception and Action Orchestrator.
- Cash/Reconciliation and Inventory/Replenishment MVP skills.
- Evaluation, policy, redaction, approval, audit, cost, and incident controls.

Build next

- Model/cost routing, explicit preference memory, offline awareness, and country-pack provenance.
- Purchasing/AP and Close/Compliance agents.
- Platform Assurance Agent.
- Carefully selected Tier 1 and Tier 2 actions.

Build later

- Payroll/Workforce Agent beyond read-only readiness.
- Customer Success/Adoption Agent after telemetry maturity.
- Voice, broader integration SDK, predictive forecasting, and multi-country expansion.

Do not build

- A custom foundation model.
- A separate AI data warehouse before a demonstrated need.
- An agent that writes directly to Prisma or bypasses services.
- Autonomous posting, payment release, stock write-off, payroll approval, statutory filing, role/entitlement change, or period close.
- One agent per page or one agent per module.
- A chatbot whose answers are not tied to tenant evidence.
- An agent marketplace before tool, security, and commercial boundaries are stable.

19. Further Questions That Would Change the Plan

1. Which five to ten paying organizations are willing to join a measured pilot, and which roles use the current Daily Digest, reconciliation, and inventory workflows most often?
2. What are the current baselines for reconciliation exception time, stockout days, emergency purchases, and manager action completion?
3. Which model providers and data-processing regions are acceptable for customer, accounting, and payroll data?
4. Which first country pack has qualified legal/accounting ownership and a production support commitment?
5. Which existing service commands are safe enough for Tier 1 drafting and Tier 2 controlled writes?
6. What package and usage model will fund inference, review, and support costs without harming SMB affordability?

20. Caveats and Assumptions

- This is a static repository and documentation assessment dated 15 July 2026. It did not query production databases, billing, customer usage, support tickets, or model-provider contracts.
- Current readiness JSON files establish that specific code and policy checks passed when generated. They are not production customer outcome data.
- The 50-paying-organization statement comes from the architecture document last updated on 23 May 2026 and may be stale.
- Weighted scores are decision aids based on repository evidence and stated strategy. They are not measured ROI forecasts.
- KPI thresholds are proposed pilot gates and should be revised after baseline telemetry and customer discovery.
- Legal, payroll, tax, and country-pack production support require qualified human review and explicit ownership.

21. Source References

Internal evidence

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Final Decision

Stoquify should become an **evidence-backed operating copilot for stock, cash, controls, and close**, not a generic AI platform. Build one trusted Command Agent, one shared Exception and Action Orchestrator, and the cash and inventory domain agents first. Reuse the platform's existing snapshots, action queues, proof trails, and service-owned controls. Earn broader autonomy only after telemetry demonstrates repeat use, measurable workflow improvement, high precision, controlled cost, and zero boundary violations.

That sequence provides the most functionality with the least architectural waste while building the moats that matter: trusted operating history, outcome-labelled workflows, OHADA country-pack provenance, low-connectivity expertise, and deep stock-to-cash-to-close continuity.